ADAPTIVE METROLOGY

Measure the change.

REFLEX

A small local model with event-triggered measurement.

Double Zero

Double Zero

A changed process needs new evidence.

First customer hypothesis

- CMP process-development and consumables teams
- Repeated calibration after equipment or material changes
- Budget: metrology and engineering time

The narrow question

- Which samples should be measured next?
- Can a local model adapt with fewer labels?
- Production release remains a separate decision

WORKING PROTOTYPE

Double Zero

Predict. Probe. Adapt.

1

Predict

Execute the trained model on observed features.

2

Probe

Request a measurement using past feedback.

3

Adapt

Update residual state after the label arrives.

GRAPE base model + event-triggered schedule. Actual local inference; public recorded data.

EXPLORATORY RESULT · DATASET 1

Double Zero

Same 88 measurements. Lower error.

Metric	Uniform GRAPE	REFLEX schedule	Difference
Measurements / 542 samples	88	**88**	**Same count**
RMSE	0.286	**0.219**	**23.4% lower**
P90 absolute error	0.393	**0.336**	**14.6% lower**

Post-holdout exploration. Full-label GRAPE achieves 0.209 RMSE with 542 measurements.

WHAT THE EXPERIMENT DOES NOT PROVE

Double Zero

The original success gate was not met.

Dataset 2

- Same 342 measurements across both schedules
- 0.841 vs 0.849 RMSE: 1.0% improvement
- 95% block-bootstrap interval includes zero

Boundaries

- CMP v1 and locked v2 candidates failed
- Event-triggered revision is exploratory
- Five-sample feedback delay removes Dataset 1 advantage

LOCAL DEPLOYMENT

Double Zero

Small enough to live beside the equipment.

Weights, scaler and schedule only. Raw sensor feature extraction is excluded.

3.4 KB

Dataset 1 model artifact

30.3 KB

Dataset 2 model artifact

NumPy

Portable runtime; no cloud call

BUSINESS HYPOTHESES · NOT VALIDATED PRICES

Double Zero

Sell a reusable model, prove the savings.

Paid validation

KRW 5–10 million
One equipment family, one measured quality target.

Integration

KRW 20–50 million
Local runtime and existing validation workflow.

Repeat deployment

Equipment/OEM licenses
Price against verified customer value.

Defensibility hypothesis: permitted change-event data, reusable adapters and validated failure boundaries.

Double Zero

The next proof is a fresh process.

Seeking a CMP data-validation partner.

Delay-aware adaptation · fresh holdout · measured engineering savings.

CONTACT

Baseline: github.com/cheonwoolee/GRAPE

Paper: arxiv.org/abs/2611.00806